

NETWORK TIME PROTOCOL

1. Project Overview:

This document highlights the design, configuration and implementation of the network time protocol built with the help of the Cisco Packet Tracer. In this project the effectiveness of using a server as a master to act as the warehouse to which other devices is shown both diagrammatically as well as logically. Connecting the client to the server, accurate time gets distributed to ensure that all devices are on the same time.

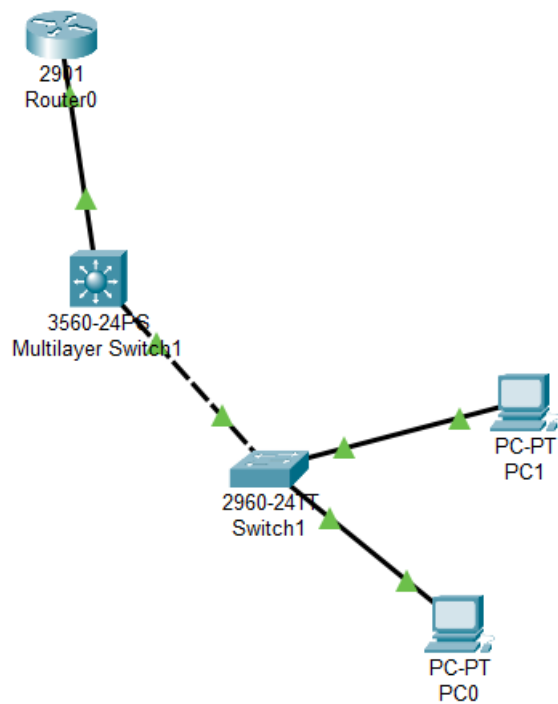
- Project Title: NetworkTimeProtocol_PathFinder
- Objective: The intention is to demonstrate how devices all the same network connected to the same network are able to request the correct time from the server through the use of NTP.
- Protocol: Network Time Protocol

2. Materials used

Item	Quantity	Model
Router	1	2901 Router
Access Switch	1	3560-24PS
Multilayered switch	1	290-24TT
PC	2	PC-PT

3. Network design and Topology:

Logical design



4.Configurations

- NTP master configs:

```
Router> en
```

```
Router# conf t
```

```
Router(config)#interface gig0/0
```

```
Router(config)# ip address 192.168.1.1 255.255.255.0
```

```
Router(config)# no shut
```

```
Router(config)# exit
```

!sets the router as the master

```
Router(config)# ntp master 1
```

- NTP client config

```
Switch> en
```

```
Switch# conf t
```

```
Switch(config)# interface fa0/1
```

```
Switch(config-if)#ip address 192.168.1.2 255.255.255.0
```

```
Switch(config-if)#no shut
```

```
Switch(config-if)#exit
```

```
!sets default gateway
```

```
Switch(config)# ip default gateway 192.168.1.1
```

```
!sets router as ntp server
```

```
Switch(config)#ntp server 192.168.1.1
```

5. IP addressing

IP addressing the end devices was not done due to the limitations of features in the Cisco Packet Tracer, the devices cannot act as clients and request time. Therefore they were only placed in the setup for completeness of the structure and nothing else, the same can be said about the access switch to which the two end devices are connected to.

6. Testing and verification

The router was selected as the ntp master

```
* invalid input detected at ... marker.  
Router#show ntp status  
Clock is synchronized, stratum 1, reference is 127.127.1.1  
nominal freq is 250.0000 Hz, actual freq is 249.9990 Hz, precision is 2**24  
reference time is EC6C59BC.000000C7 (20:10:36.199 UTC Tue Oct 14 2025)  
clock offset is 0.00 msec, root delay is 0.00 msec  
root dispersion is 0.00 msec, peer dispersion is 0.47 msec.  
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is - 0.000001193 s/s system  
interval is 6, last update was 45 sec ago.  
Router#
```

The router was set to a specific time and date so that the offset is not very large

```
Router#show clock
20:16:22.988 UTC Tue Oct 14 2025
Router#
```

The ntp associations

```
Router#show ntp associations
```

```
address      ref clock      st  when    poll    reach  delay      offset
*~127.127.1.1 .LOCL.         0   15      64     377    0.00      0.00
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
Router#
```

The switch was set as the client and had to request the correct time from the router

```
Switch>en
Switch#show ntp associations

address      ref clock      st  when    poll    reach  delay      offset
*~192.168.1.1 127.127.1.1    1   13      16     377    0.00      0.00
* sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
Switch#
```

(NOTE: The * indicates that the client and the server is synchronized)

The NTP status of the client

```
Switch#show ntp status
Clock is synchronized, stratum 16, reference is 192.168.1.1
nominal freq is 250.0000 Hz, actual freq is 249.9990 Hz, precision is 2**24
reference time is EC6C80EC.000001B4 (22:57:48.436 UTC Tue Oct 14 2025)
clock offset is 0.00 msec, root delay is 0.00 msec
root dispersion is 10.24 msec, peer dispersion is 0.12 msec.
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is - 0.0000001193 s/s system
interval is 4, last update was 4 sec ago.
Switch#
```

It can further be seen that the client synchronizes with the server, and it is at stratum 16 while the server itself is the master and is at stratum 1. It can further be noticed that both the server and the client have the same date as 14 October 2025.

7. Advantage:

- If all devices in the network system are synchronized with the master ntp 1, security issues decrease, and it offers high accuracy and scalability.

- Broad compatibility as NTP is supported by most OS.
- Accuracy in time and date ensures that troubleshooting is not too much of an issue.

8. Disadvantages:

- If the master device is down the entire system may get on different times and dates.
- It may be vulnerable to attacks as a hacker may set an offset time and cause a delay in the delivery of the packets.
- It gets quite complex when used in big organizations.

9. Conclusion:

It can now be concluded that the successful setup of one device to act as the NTP server for all devices on the network system ensures that all the devices get accurate and consistent time synchronization across the network devices. Getting all the devices to be synchronized eliminates quite a few issues such as logging in, troubleshooting, authentication and enhances how the entire system works. It can now be said that the use of the Network Time Protocol is quite crucial as it demonstrates why its important for all devices to be in sync allowing for smoother communication and management in the system.